DRAFT v0.1 - anonymised for double-blind review. Red [bracketed] items are NOT yet measured or decided. Remove this banner before submission.

AgriVision: A Low-Cost IoT Rover with Two-Stage Computer Vision for Autonomous Crop Health Monitoring

Anonymous Authors
Affiliation withheld for double-blind review

***Abstract*—Smallholder farmers can rarely inspect every plant in a field often enough to catch disease, fire or water stress early. This paper presents the design of AgriVision, a low-cost agricultural rover that combines ESP32-based multi-sensor telemetry (soil moisture, rain, ambient light, five-channel flame and temperature/humidity), ultrasonic obstacle avoidance and remote driving through a web dashboard with a pan-tilt camera feeding a two-stage vision pipeline. A lightweight detector first localises plants; a compact classifier then judges leaf health and raises a dashboard alert carrying time, class, confidence, location and a severity estimate. The sensing, driving, communication and dashboard layers are implemented; the vision pipeline, autonomous plant approach and cellular link are being integrated. We describe the architecture, state each module's implementation status explicitly, and define a plant-level evaluation protocol that includes a cross-domain test on field images. [Replace with one sentence of headline measured results]**

***Keywords*—precision agriculture, IoT, ESP32, plant disease detection, YOLO, MobileNetV3, mobile robot, edge computing**

I. INTRODUCTION

Crop losses are frequently caused by problems that are visible early but only if someone is looking: leaf disease, pest damage, dry soil, flooding after rain, or a small fire. Smallholder farmers often cannot walk an entire field several times a day, and fixed sensor nodes observe only the point where they are planted. A mobile platform that carries both environmental sensors and a camera can close this gap, provided it is cheap enough to be realistic for small farms and reports to the farmer remotely.

Most published plant-disease classifiers are trained and tested on single-leaf images with plain backgrounds, and their accuracy is known to fall on images taken in real fields [1], [3]. Most low-cost farm robots, in turn, stop at telemetry and do not interpret the camera feed. AgriVision targets the space between: an inexpensive rover whose sensing stack is working end to end, with a vision pipeline designed and evaluated with the field domain gap in mind.

The contributions of this work are:

- A low-cost rover architecture with three ESP32 nodes (driver, fire, camera), a Node.js server and a web dashboard, with measured link-quality telemetry.
- A two-stage vision design (plant detection, then leaf-health classification) driven by a pan-tilt camera, with a structured alert record for the farmer.
- An evaluation protocol that splits data by plant rather than by image and reports in-domain and cross-domain accuracy, including images from the rover's own camera.
- An explicit module-by-module implementation status table, so that built, in-progress and planned parts are not confused.

II. RELATED WORK

A) Deep learning for plant disease

Mohanty et al. trained deep networks on the PlantVillage collection [1], [2] and reported high accuracy on held-out lab images, while noting a large drop when the same models saw images from other sources. Ferentinos compared several architectures on a larger multi-crop set [4]. PlantDoc was introduced as a field-image benchmark for visual plant disease detection precisely because lab-trained models transfer poorly [3]. Kamilaris and Prenafeta-Boldú surveyed deep learning across agriculture and highlighted dataset scarcity and generalisation as open problems [5].

B) Lightweight detectors and classifiers

Single-stage detectors from the YOLO family [6], [7] run in real time on modest hardware, and MobileNetV3 [8] provides compact image classifiers suited to edge deployment. We use them as building blocks rather than proposing new architectures.

C) IoT monitoring and agricultural robots

[ADD 4-6 verified references on IoT field monitoring, low-cost agricultural robots and LoRa/cellular links in farms; see the literature-search guide] Prior systems typically provide either fixed-point sensing or a mobile platform without image interpretation; the gap this work addresses is a low-cost mobile platform that joins multi-sensor telemetry with two-stage plant-health vision under an honest, field-oriented evaluation.

III. SYSTEM ARCHITECTURE

A) Overview

AgriVision consists of a rover carrying three microcontroller nodes, a link to a server, and a browser dashboard (Fig. 1). A driver node (ESP32) controls the motors through an L298N H-bridge and reads the soil-moisture, rain and ambient-light sensors, a temperature/humidity sensor and ultrasonic range finders. A separate fire node (ESP32) reads a five-channel flame sensor, and a camera node (ESP32-CAM) on a two-servo pan-tilt mount supplies images.

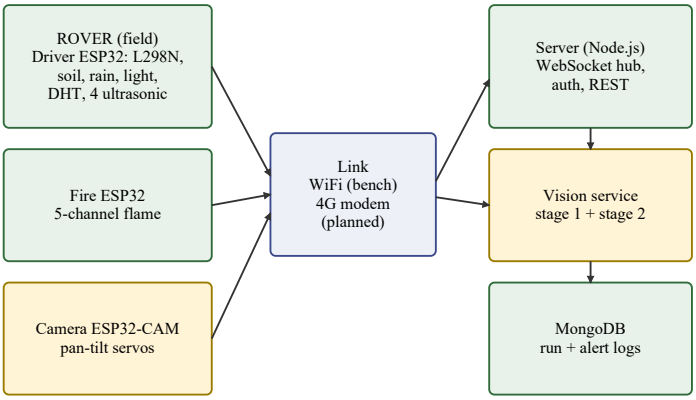


Fig. 1. System architecture. Amber blocks are in progress or planned.

B) Sensing and actuation

Soil moisture and rain are read on analog inputs and mapped to percentages using calibration constants measured in dry air and in water. A light sensor switches the front and rear lamps automatically with a 1.5 s hold to avoid flicker. The flame

channels use a 150 ms assert and 600 ms clear debounce so that flicker does not chatter. Obstacle sensing uses an ultrasonic range finder on a scanning servo; the final design uses four ultrasonic sensors, the front one on the servo, triggered one at a time to avoid crosstalk, with 5 V echo lines level-shifted to the 3.3 V ESP32 inputs.

C) Software and communication

Nodes connect over WiFi to a Node.js server through WebSockets and register by device identifier. The dashboard (React) sends drive commands at a 100 ms cadence; the rover stops itself if commands cease for 500 ms (failsafe). Every 2 s the dashboard sends a probe that travels dashboard-server-rover and back, from which round-trip time, loss and the rover's WiFi signal strength are computed. Accounts are authenticated, and run history is stored in MongoDB without sitting on the driving path, so a database outage does not affect control. For farm deployment the WiFi link is to be replaced by a 4G modem and a cloud-hosted server, with the camera sending event-triggered snapshots rather than continuous video to limit data use.

D) Implementation status

Table I states what exists today. This table is revised at each draft.

TABLE I. MODULE STATUS

Module	Status
Motor drive (L298N), dashboard control, failsafe	Implemented
Soil, rain, light sensing; auto lamps	Implemented
5-channel flame sensing (separate node)	Implemented
Link-quality probes (RTT, loss, RSSI)	Implemented
Auth, run history (MongoDB)	Implemented
Scanning ultrasonic + reactive avoidance (firmware)	Implemented **[field test]**
4 ultrasonic sensors; DHT sensor	In progress
Pan-tilt camera; plant detector; leaf classifier	In progress
Plant approach; severity %; plant location	Planned
4G modem + cloud server	Planned
Pesticide/treatment actuator	Future work

IV. VISION PIPELINE

A) Camera node

An ESP32-CAM serves JPEG snapshots over HTTP. Two hobby servos give pan and tilt so that the rover can scan a row without turning the chassis. Servos are powered from a separate 5 V supply with a bulk capacitor and a common ground, because the camera module browns out when sharing a weak supply.

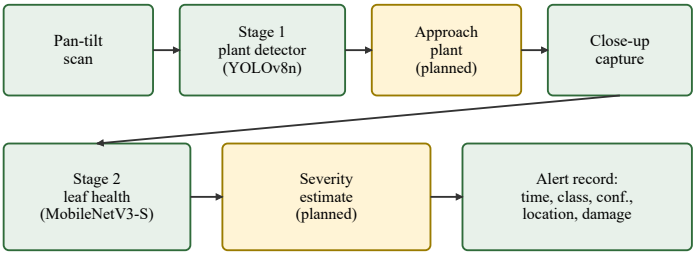


Fig. 2. Two-stage vision pipeline. Amber blocks are planned.

B) Stage 1: plant detection

A single-class YOLOv8n detector [7] answers whether a plant is in view and where. Training combines plant-positive images with hard negatives (people, soil, pots, buildings) because an earlier detector trained without negatives labelled people as plants. The model is exported to a lightweight runtime for inference. **[training set size, epochs, image size]**

C) Stage 2: leaf-health classification

Close-up crops of the detected plant are classified by a MobileNetV3-Small network [8] fine-tuned from ImageNet weights. We first target tomato, with a binary healthy/diseased output and a per-disease output, and report both. Class names follow the public tomato categories [2]. **[number of classes, augmentation, optimiser]**

D) Alert record and severity

An unhealthy decision creates a record with timestamp, crop, predicted class, confidence, thumbnail, rover position and a severity estimate. Severity is planned as the fraction of leaf pixels flagged as lesion by a colour or segmentation step, which is a proxy for damage and not a clinical measure. Rover position is planned from dead reckoning or GPS. **[final severity method]**

V. EVALUATION PROTOCOL

Datasets. Public tomato images from PlantVillage [2] provide the in-domain set; PlantDoc [3] provides field images for a cross-domain test; and a set photographed in a real tomato field with the rover's ESP32-CAM provides the deployment test. Each field image records plant ID, date, time and weather.

Splitting. Splits are made by plant (or capture session), never by image, so near-duplicate photographs of one leaf cannot appear in both training and test sets. Labels for field images are verified by an agriculture expert **[name role only, no identity]**

Metrics. For stage 1: precision, recall, mAP@0.5, mAP@0.5:0.95 and latency. For stage 2: accuracy, macro-F1, per-class recall and the confusion matrix, reported as mean and standard deviation over **[3]** seeds. For the system: sensor-to-dashboard latency, packet loss, WiFi signal strength versus distance, camera frame time, and battery runtime; the same link tests are repeated over a 4G path **[hotspot or modem]**.

VI. RESULTS

No measured results are included in this draft. The tables below show the quantities that will be reported.

TABLE II. STAGE 1 PLANT DETECTOR

Model	P	R	mAP50	mAP50-95	ms
YOLOv8n	[-]	[-]	[-]	[-]	[-]

TABLE III. STAGE 2 LEAF-HEALTH CLASSIFIER

Test set	Acc.	Macro-F1
PlantVillage tomato (in-domain)	[-]	[-]
PlantDoc tomato (cross-domain)	[-]	[-]
Own field set, ESP32-CAM	[-]	[-]

TABLE IV. SYSTEM LINK MEASUREMENTS

Metric	WiFi	4G
Median RTT (ms)	[-]	[-]
Packet loss (%)	[-]	[-]
Snapshot delivery (s)	[-]	[-]
Distance tested (m)	[-]	[-]

[Discussion: compare in-domain and cross-domain accuracy; explain failures with example images; state what drove the gap]

VII. LIMITATIONS AND FUTURE WORK

Ultrasonic sensing reflects poorly from soft foliage, so avoidance is reliable for hard obstacles but not guaranteed for dense plants; full path planning would need GPS waypoints or vision-based row following. The temperature/humidity sensor used in the prototype is low-cost and not weatherproof; a field version would use a more accurate sealed sensor and a soil-temperature probe. A classifier trained mostly on lab images may degrade in sunlight and mud, which is why the cross-domain test is reported. Treatment actuators (spraying)

raise safety questions and are left to future work. Finally, one crop is evaluated; multi-crop generalisation is not claimed.

VIII. CONCLUSION

AgriVision demonstrates a low-cost rover whose sensing, control and dashboard layers are implemented, and defines a two-stage vision pipeline and an evaluation protocol built around the field domain gap. **[Replace with 2 sentences on measured findings]** Remaining work covers the camera integration, autonomous approach, cellular link and field trials.

REFERENCES

[1] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Front. Plant Sci.*, vol. 7, Art. 1419, 2016.

[2] D. P. Hughes and M. Salathé, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," arXiv:1511.08060, 2015.

[3] D. Singh *et al.*, "PlantDoc: A dataset for visual plant disease detection," in *Proc. 7th ACM IKDD CoDS and 25th COMAD*, 2020, pp. 249–253.

[4] K. P. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Comput. Electron. Agric.*, vol. 145, pp. 311–318, 2018.

[5] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep learning in agriculture: A survey," *Comput. Electron. Agric.*, vol. 147, pp. 70–90, 2018.

[6] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE CVPR*, 2016, pp. 779–788.

[7] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," software, 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[8] A. Howard *et al.*, "Searching for MobileNetV3," in *Proc. IEEE/CVF ICCV*, 2019, pp. 1314–1324.

[Verify every reference against the original source before submission; add the IoT/robotics references]